LAB

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PROGRAMMING CONCEPTS

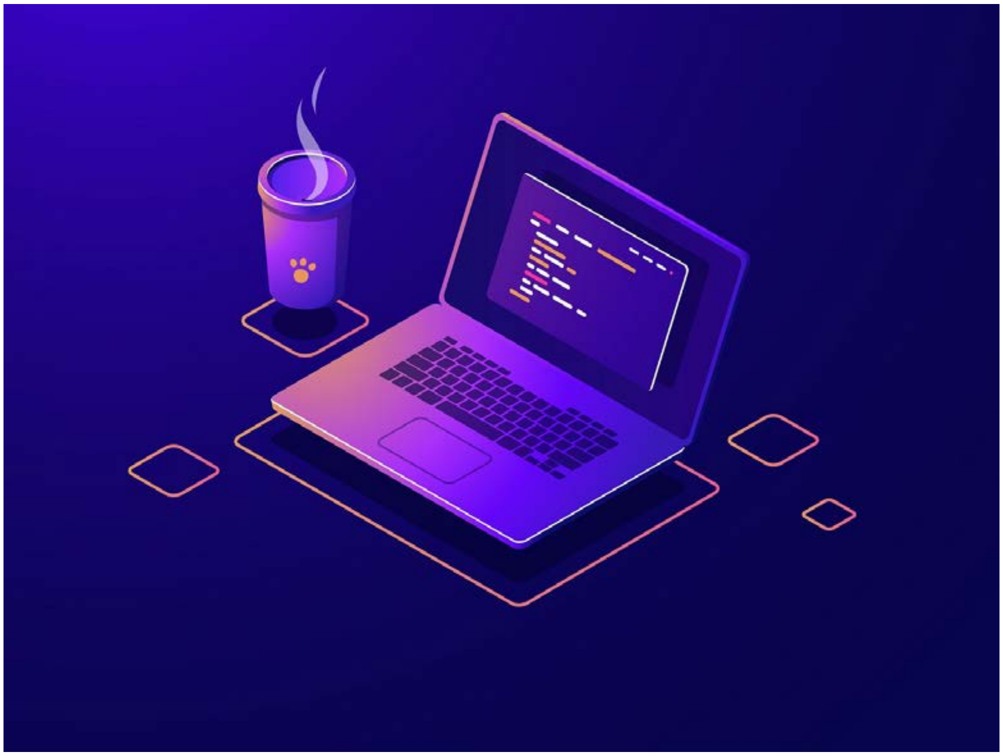
Welcome to Programming!

LEARNING OBJECTIVES

At the end of this lab, you should be able to:

- Run programs using Visual Studio Code.
- Write simple programs using input and output statements.

WELCOME!



Welcome to programming, the field that changed the world!

If you do not believe that, close your eyes and imagine how the world would be without the many applications we daily use online and on our electronic devices!

You should surely be proud that you are on your first step to be among those who develop applications to change the world.

HELLO WORLD!

For every programming language you will learn, there is always a very first program that will introduce you to the language and the world of programming: “Hello World” program!
The Hello World program is the simplest program you can ever write. When you run it, it just displays one message on the screen. Guess which message?

```
print("Hello World")
```

After you run this program (yes, it is a program that has just one command), let's try to change the double quotes to single quotes. What happens?

```
print('Hello World')
```

Can you mix both? What do you get when you try to run the following statement?

```
print("Hello World')
```

GETTING INPUT

Now, let's write a program that greets you by your name when you run it! It should first ask you about your name, and then display the greeting message. A Python program interacts with its user using the `input` statement. Here is an example.

```
nm = input("What is your name?")
```

With the above statement, a message `"What is your name?"` will be displayed, then the program will wait until the user enters his/her name. Once the user does so, the program stores it in a "container" called `nm`. We call such a container a *variable*. A variable is a way of storing information in a computer program. Think of a variable like a container and the name of the variable (`nm` in our case) as the label on the container. We can then use such a name to show us what is inside the container, i.e., refer to the value stored in it.

Now the program can greet you by name.

```
print("Hello ", nm)
```

It can even do so in a better way.

```
print("Hello ", nm)

print("Welcome to our class!")
```

EXERCISES

(1) Modify the above program to also ask you for your major in addition to your name, then telling you (after the greeting message) that your major (by its name) is fun!

(2) Extend the program in **(1)** to also ask you for your age in addition to your name and major, then print your age after the greeting and major messages.